

Virtual International Seminar on Theoretical Advancements

Nonadiabatic excited-state dynamics in nanoscale TiO_2 : from size and environment to fluxional nuclear motion

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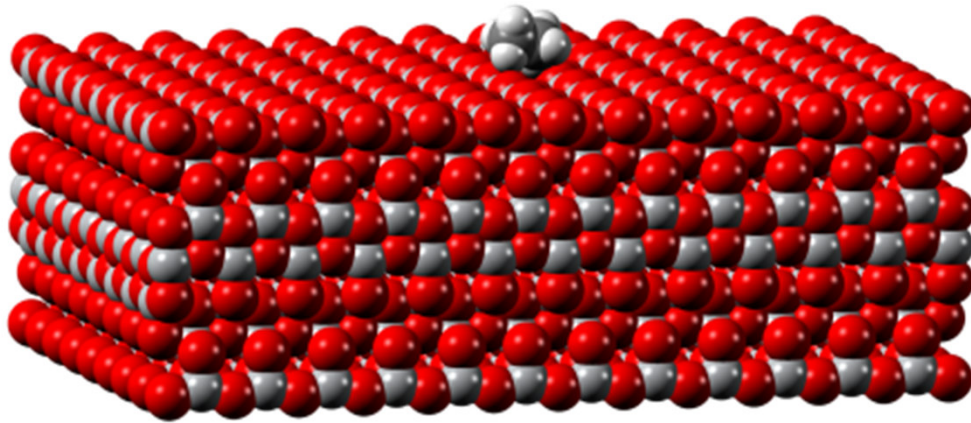


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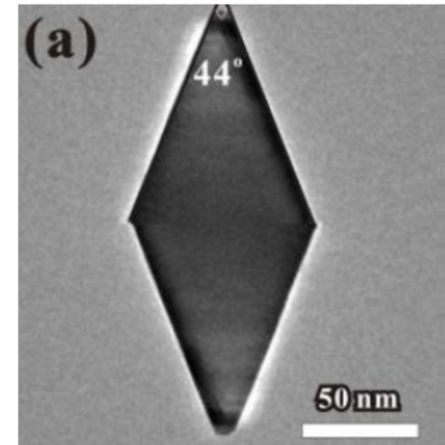
UB University
at Buffalo

TiO_2 is a benchmark photoactive oxide

used in photocatalysis, environmental remediation, or solar-energy conversion



Extended/surface models
simplified, periodic... idealized

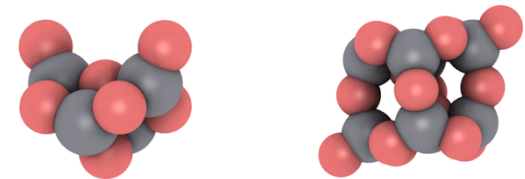


Lee et al., ACS Omega **2018**, 3, 10225

Real **nanoparticles**
realistic, but structurally complex

Nanoclusters: Controlled finite models

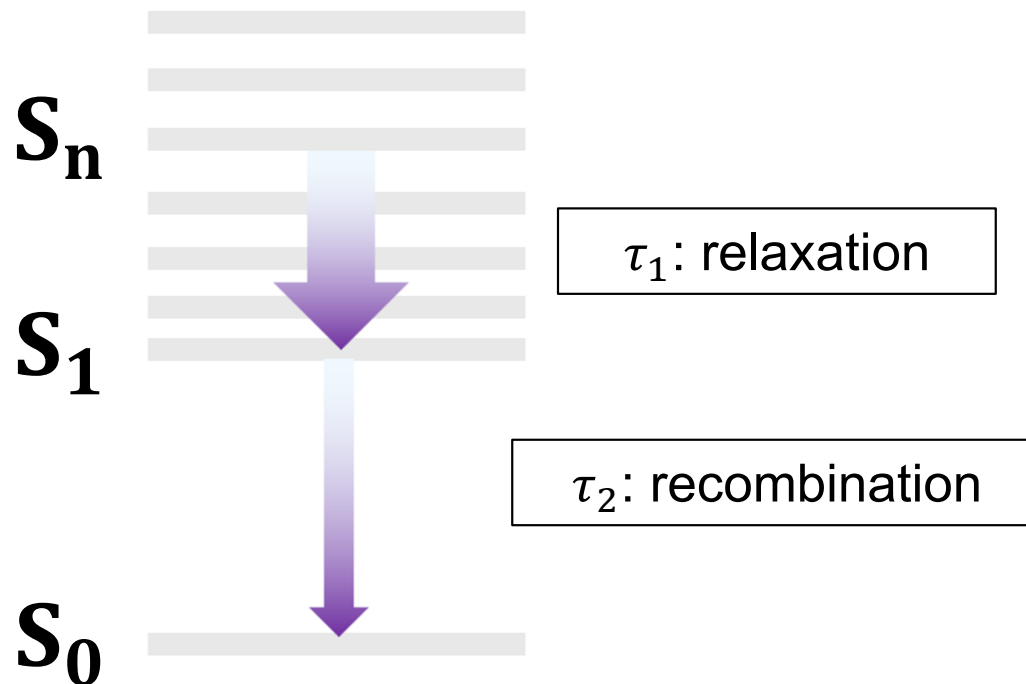
- known size and structure
- size and finite-temperature motion effects?



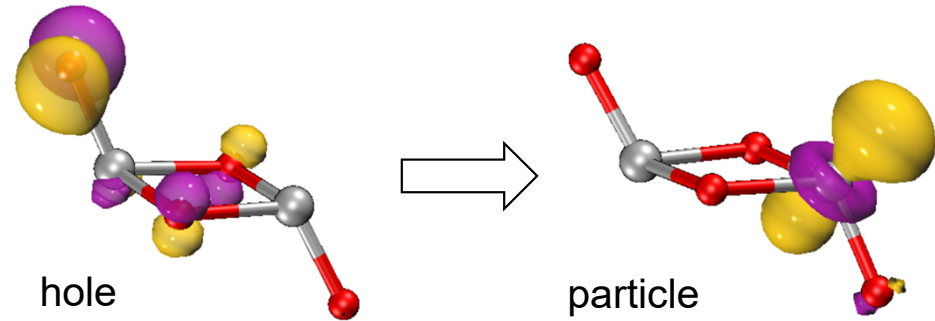
light absorption

- e^- / h^+ generation
- charge separation
- surface reactions

-

→ sequential excited-state decay:

Electronic picture:

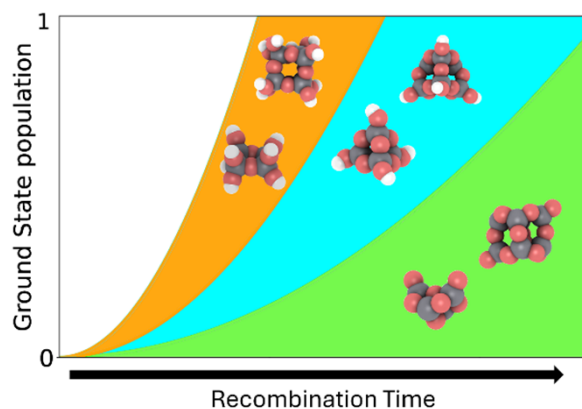
 $\text{O}_{2p} \rightarrow \text{Ti}_{3d}$ excitation

What controls these timescales in TiO_2 nanoclusters?

Three ways to reshape the excited-state landscape

Chemical environment:
bare $\rightarrow$ hydrated

Hydroxylation changes
excitation energies and
NACs

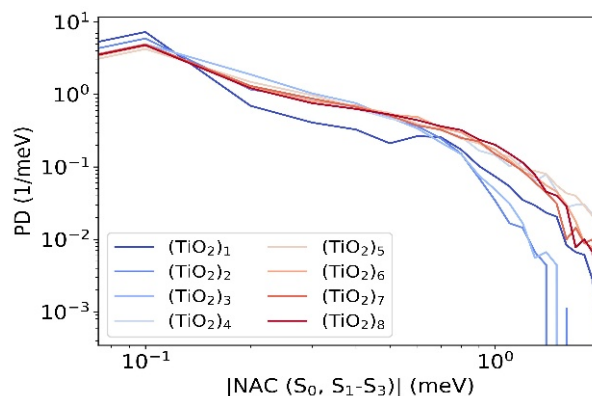


Faster recombination

JPCC 2025, 129 (3), 1806-1823

Size:
small $\rightarrow$ large

Cluster size changes
DOS, gaps, and state
mixing

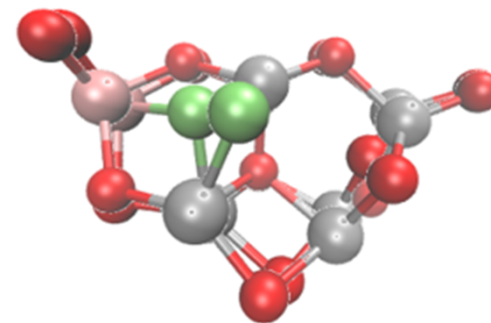


**Non-trivial size
trends**

JACS 2025, 147(44), 40900-40909

Nuclear motion:
rigid $\leftrightarrow$ fluxional

Fluxionality changes the
sampled excited-state
landscape

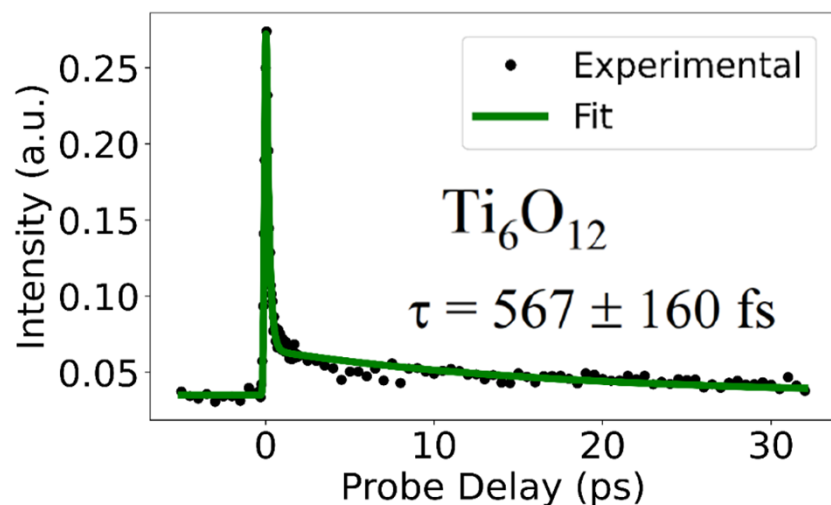


**Altered relaxation and
recombination**

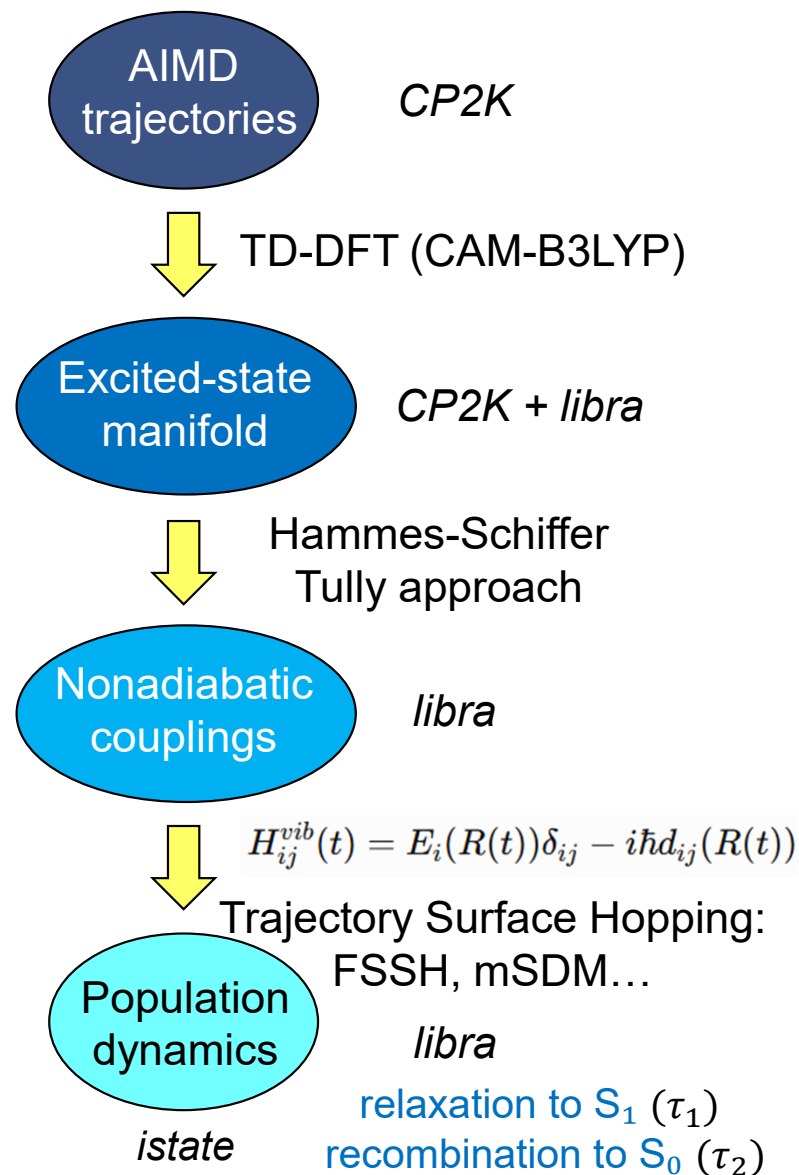
JPCL 2026

Strategy: experiment + mixed quantum–classical dynamics

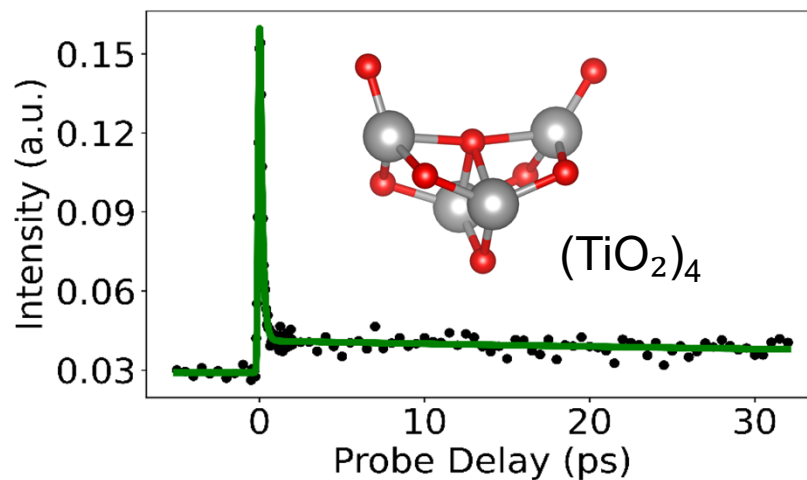
$(\text{TiO}_2)_n$, $n = 1-8$ femtosecond
pump–probe spectroscopy
observed lifetimes: τ_1 , τ_2



Scott G. Sayres, **ASU**



Experiment



$$\tau_1 = 152 \pm 9 \text{ fs}$$

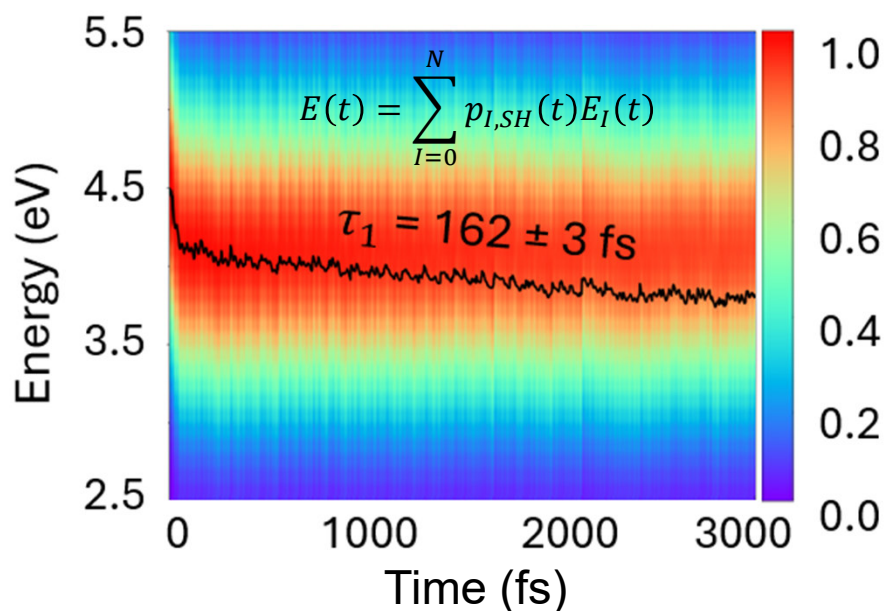
$$\tau_2 = 101.3 \pm 33.0 \text{ ps}$$

Two dynamical regimes require two observables:

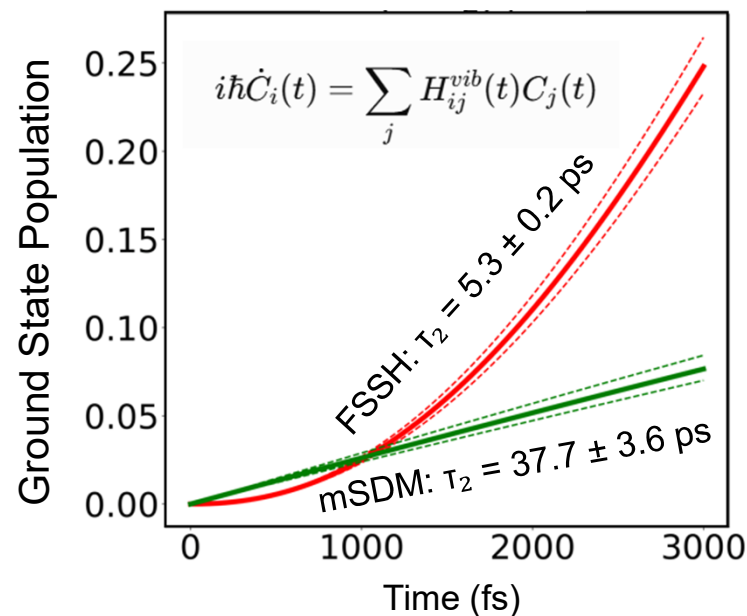
τ_1 : average excitation-energy decay

τ_2 : S_0 population recovery

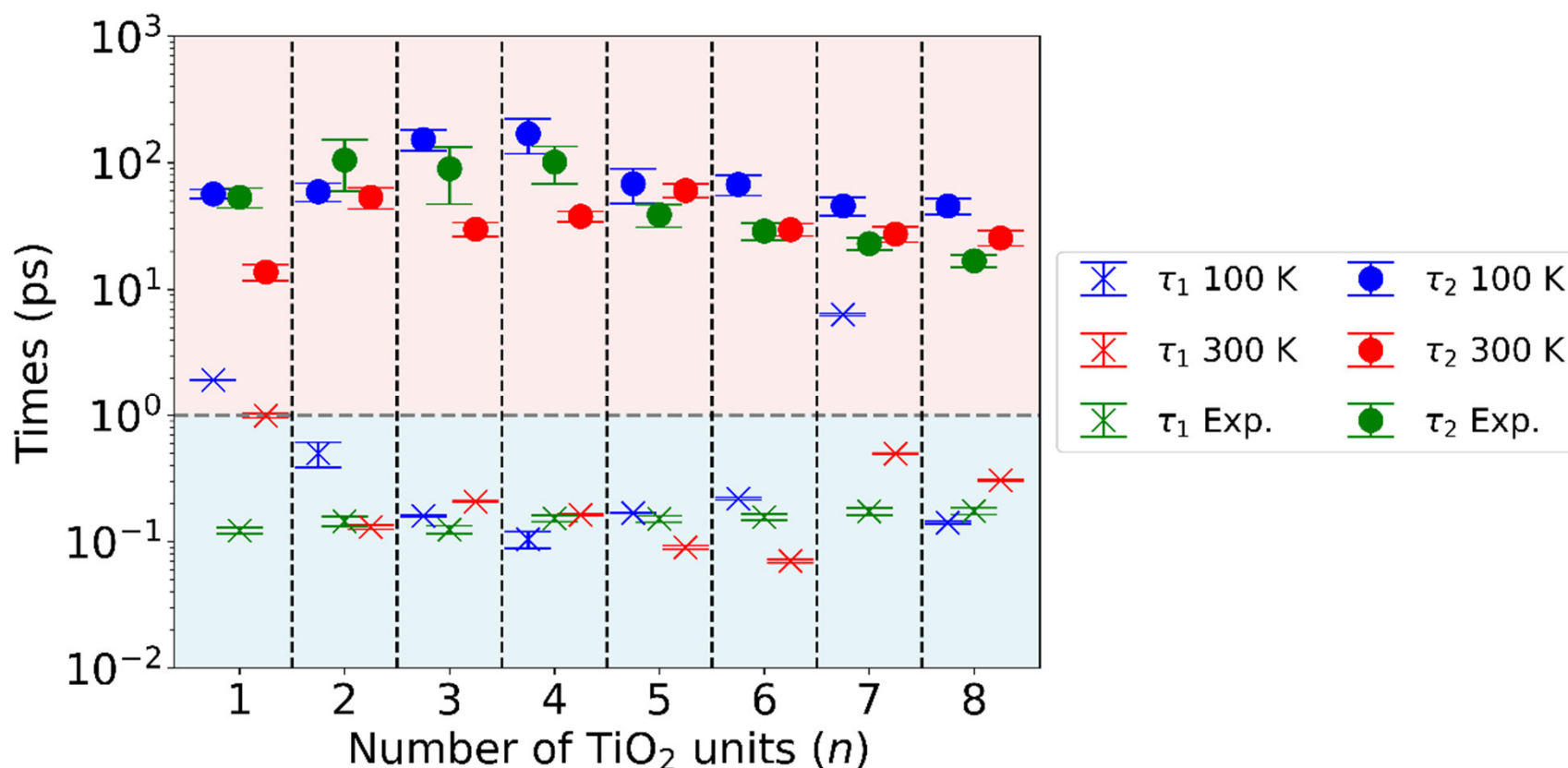
Relaxation observable



Recombination observable

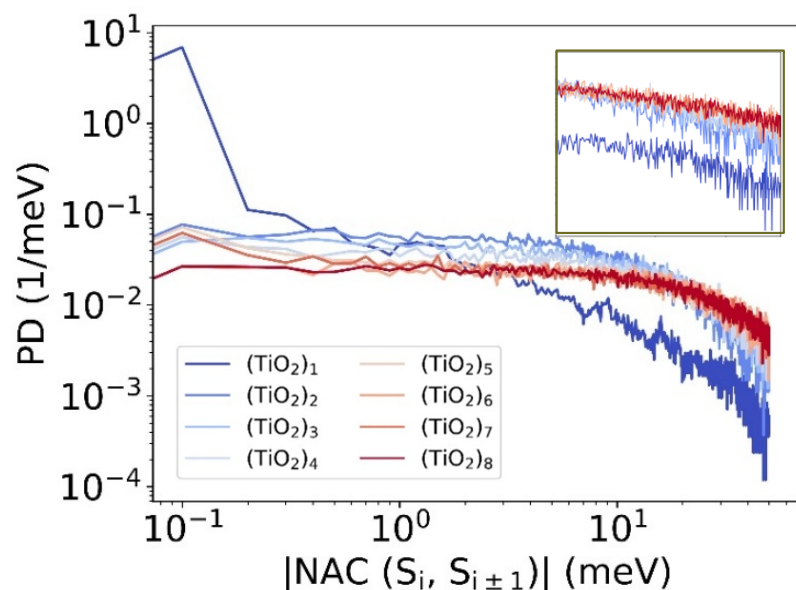
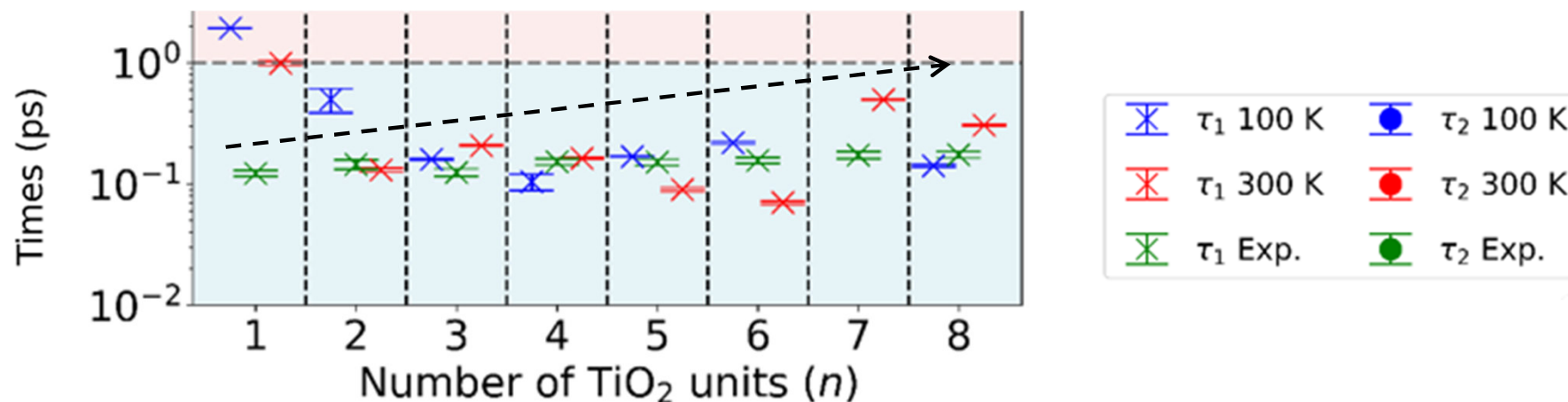


Experiment and NA-MD reveal size-dependent lifetimes



Two separated regimes: $\tau_1 \ll \tau_2$; Size dependence is non-trivial, especially for τ_2
Higher temperature accelerates recombination
Good agreement with pump-probe lifetimes

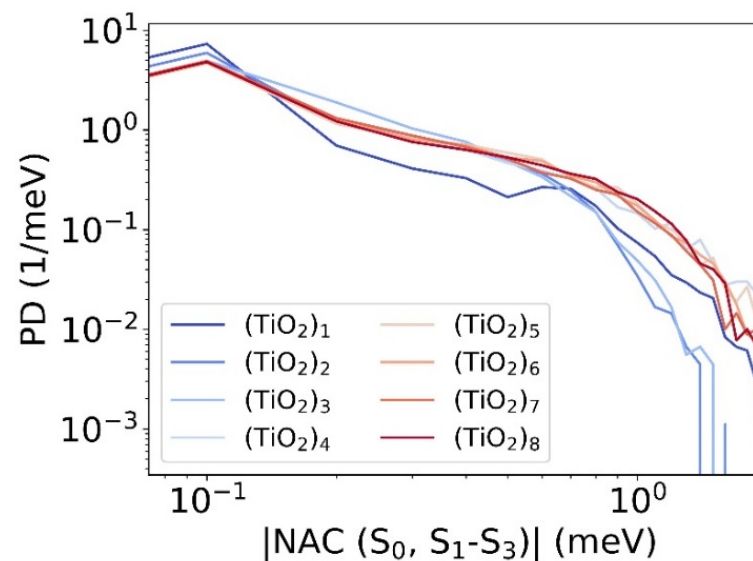
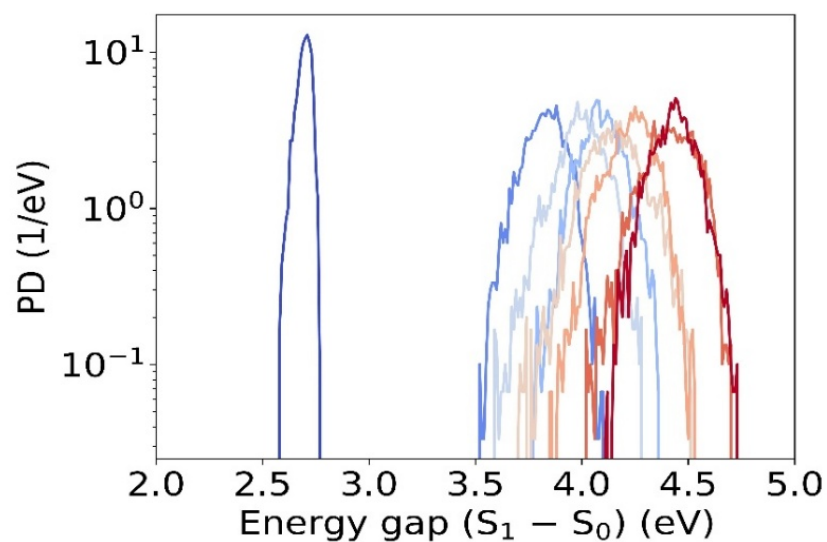
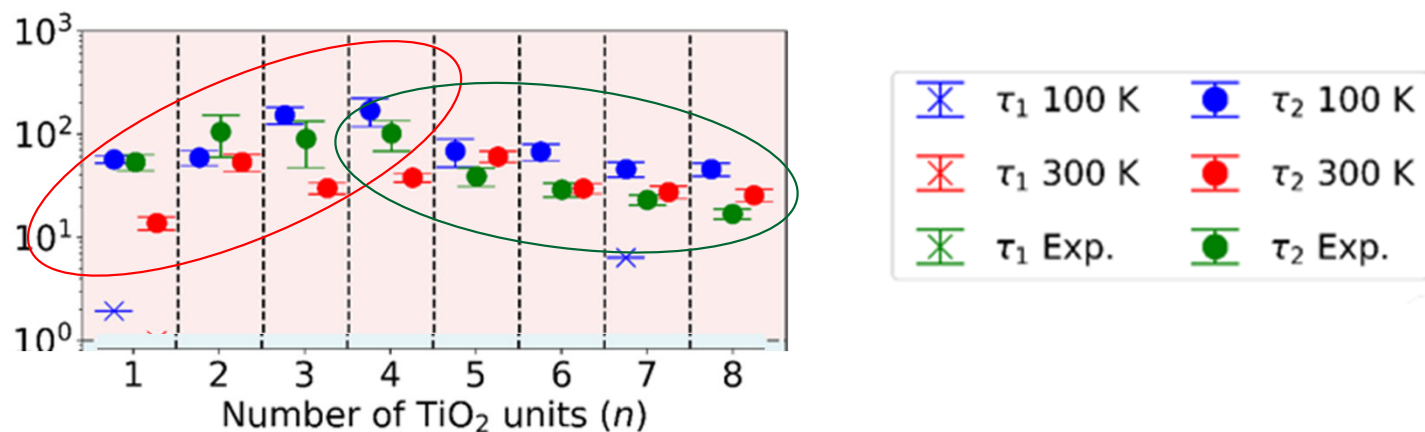
Microscopic origin of the size-dependent lifetimes (I)



Relaxation: motion within excited-state manifold.

Denser manifolds and adjacent-state NACs promote population transfer and spreading in larger systems

Microscopic origin of the size-dependent lifetimes (II)

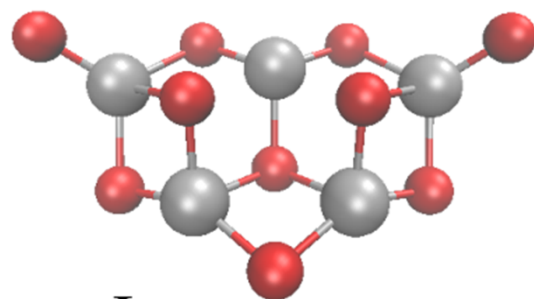
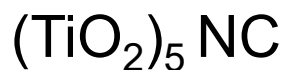


Recombination: coupling back to S_0 .

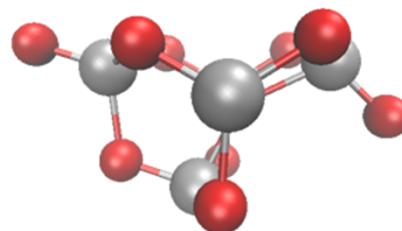
Larger gaps slow recombination; larger S_0/S_1 NACs accelerate it.

Beyond size: which geometries define the landscape?

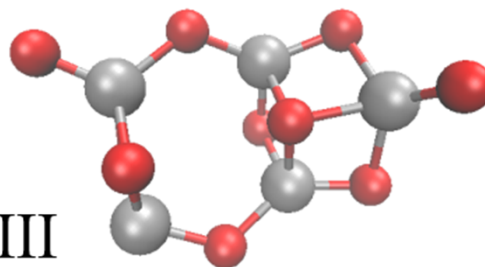
Nanoclusters are not always single static structures!
At finite temperature: shallow PESs can sample multiple motifs.



Isomer I



Isomer II

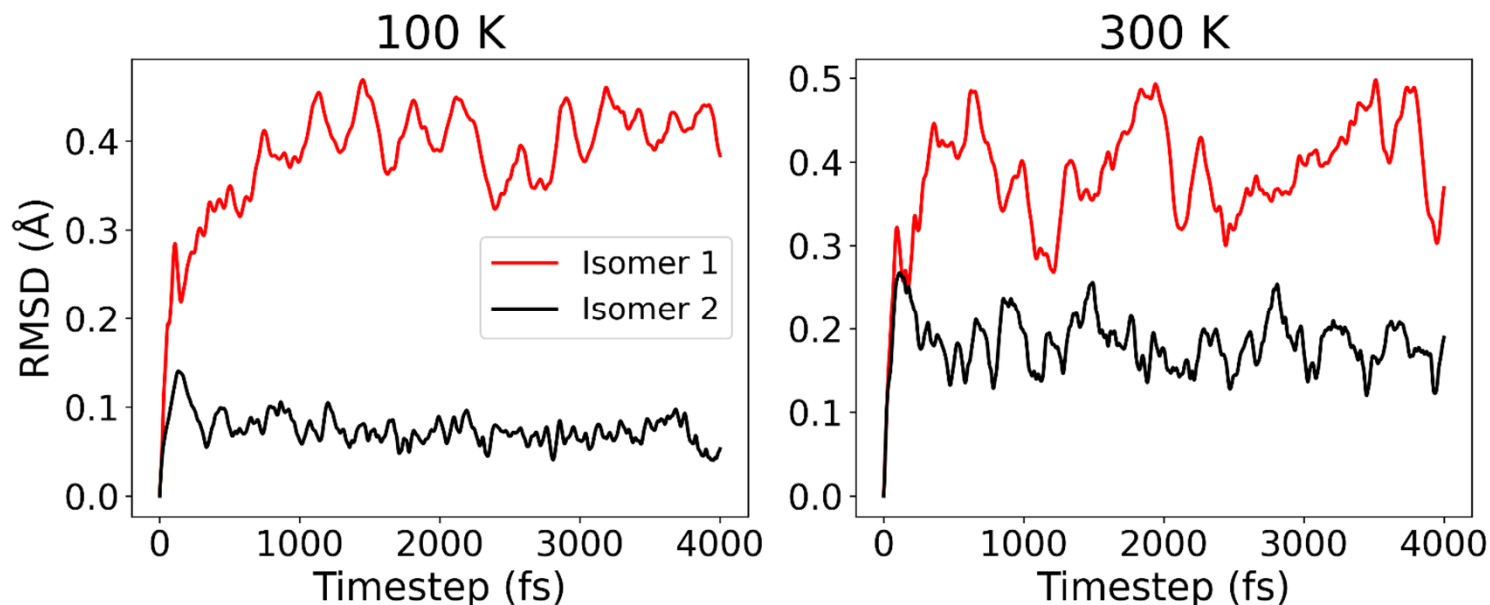


Isomer III

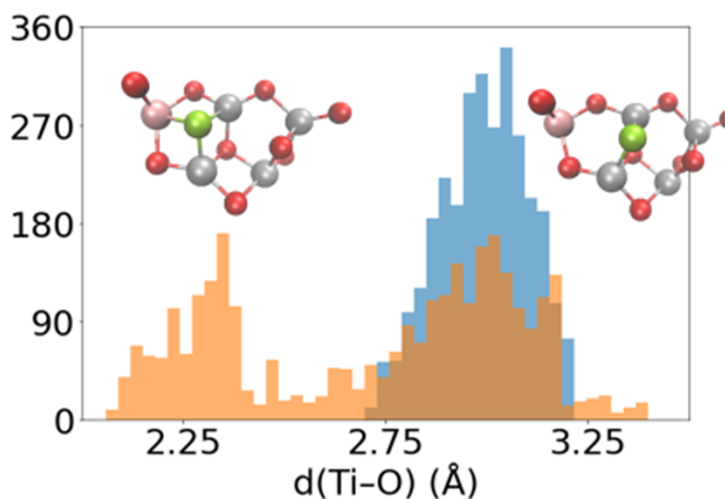
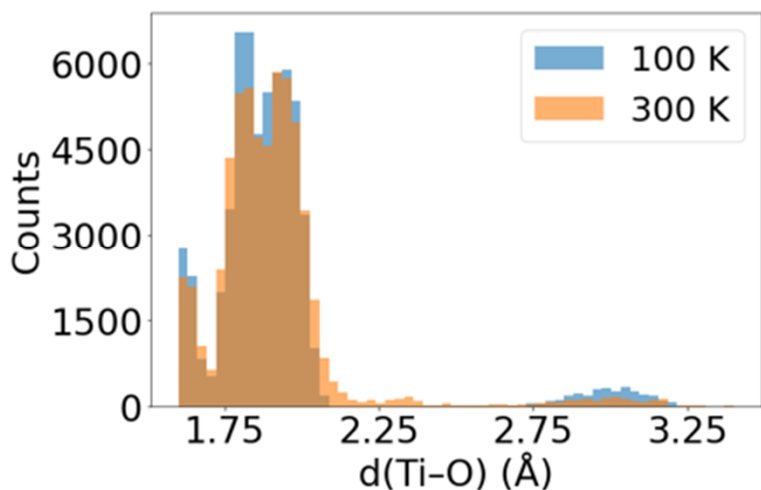
Isomer I (0.00 eV)
Isomer II (+0.02 eV)
Isomer III (+0.08 eV)

Can thermally activated structural motion reshape excitation energies, NACs, and excited-state decay pathways?

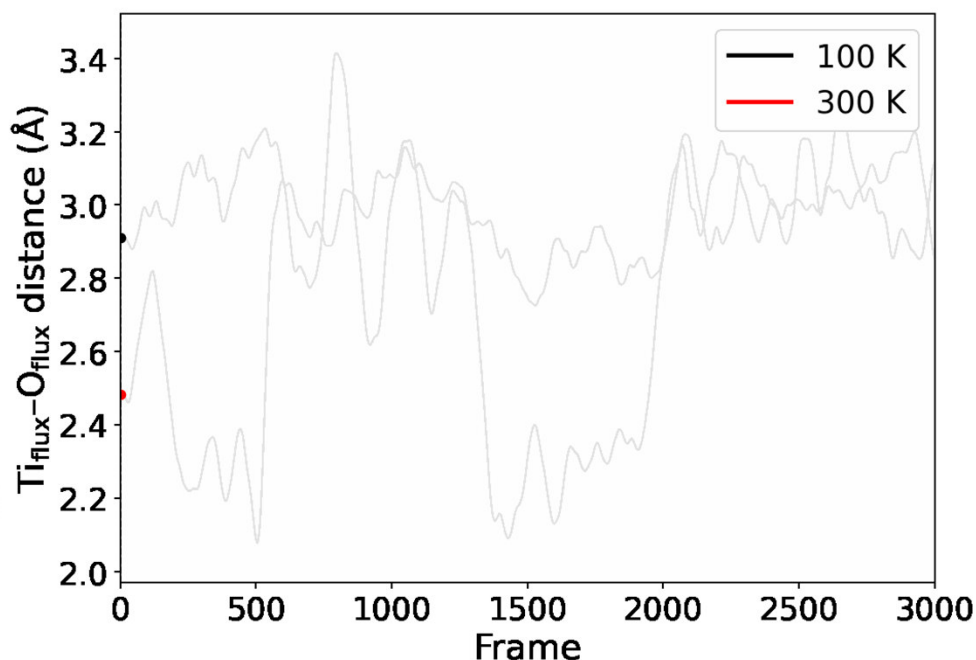
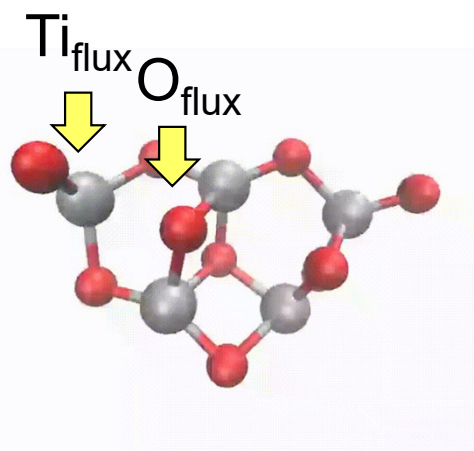
Thermally activated local fluxionality



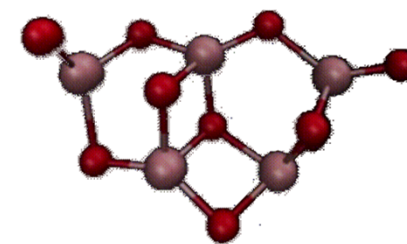
Isomer I samples a broader configurational space than rigid isomer II.



Not a random global distortion!
One $\text{Ti}_{\text{flux}}-\text{O}_{\text{flux}}$ pair switches between coordinated and detached regimes

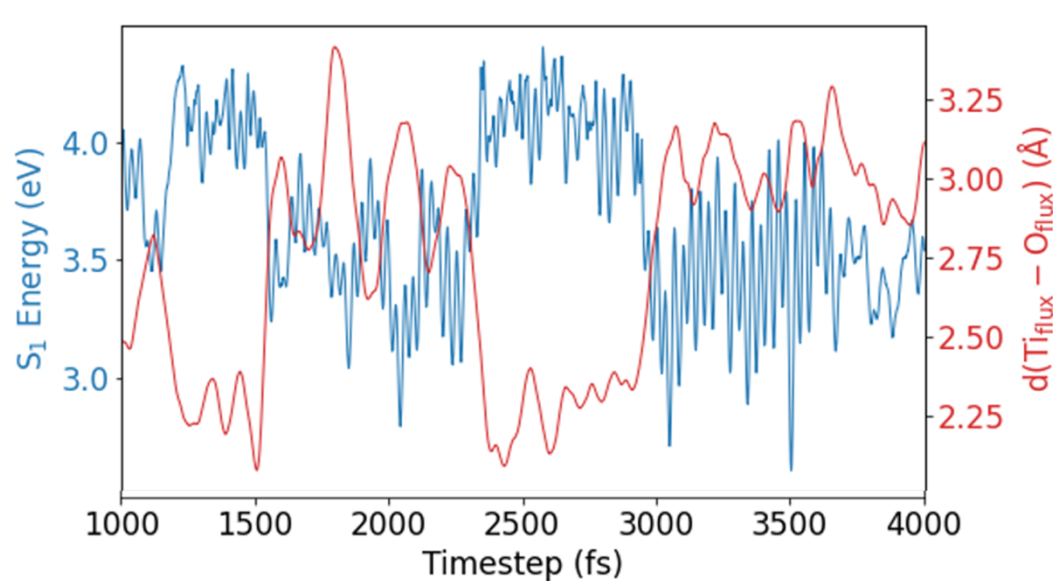


Part of a real
collective
motion:

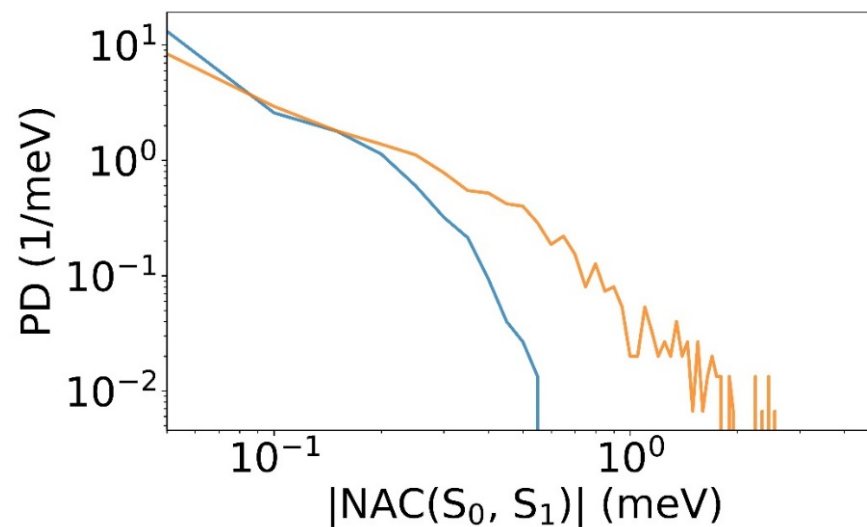
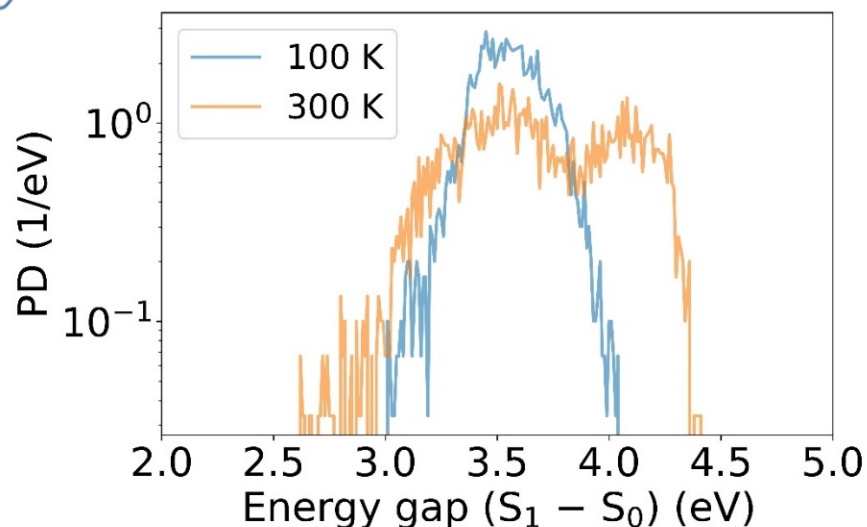


Strachan-v

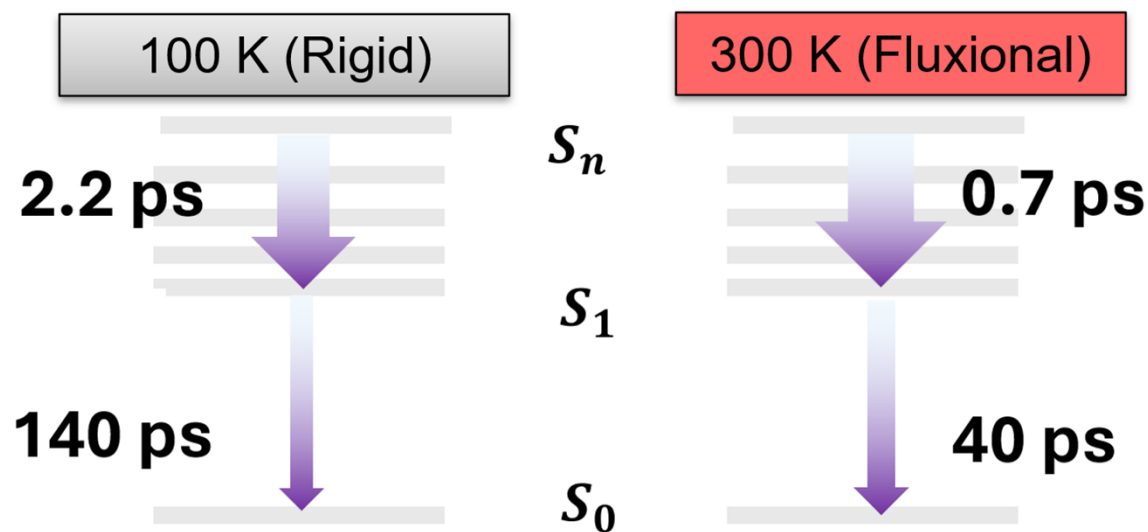
Fluxionality reshapes the low-energy excited-state landscape



Anti-correlation: elongation of
 $\text{Ti}_{\text{flux}}-\text{O}_{\text{flux}}$ lowers the S_1
excitation energy



Bimodal S_1-S_0 gap distribution and a stronger high-NAC tail at 300 K

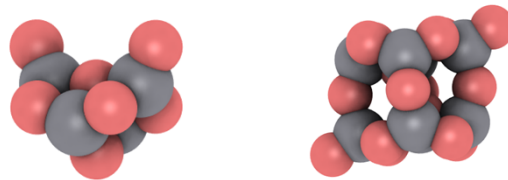


Relaxation to S_1 is faster at 300 K because the bright state lies closer to the S_1 destination

$S_1 \rightarrow S_0$ recombination accelerated because 300 K samples stronger-coupling geometries

Take-home message:

Size defines the electronic landscape; nuclear motion determines how it is sampled. Together, they control excited-state lifetimes.



- Size effect: DOS · gaps · state mixing · NACs → non-trivial relaxation/recombination trends
- Fluxionality effect: finite-temperature Ti–O motion → altered excitation energies and faster decay pathways
- TiO₂ nanoclusters provide atomically precise platforms to resolve these effects.

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